

OPERATING SYSTEMS CONCEPT – LAB

EXP – 07: PRODUCER CONSUMER PROBLEM SOLUTION USING SEMAPHORE

```
Activities Terminal Apr 20 17:39
rathnavelu06@localhost:~/rathna/producerConsumer — /usr/bin/vim pc.c

#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in=0;
int out=0;
sem_t empty;
sem_t full;
sem_t mutex;

void *producer(void *arg){
    int item;
    for(int i=0; i<BUFFER_SIZE; i++){
        item=i;
        sem_wait(&empty);
        sem_wait(&mutex);
        buffer[in]=item;
        printf("Producer: %d\n", item);
        in=(in+1)%BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&full);
    }
}

void *consumer(void *arg){
    int item;
    for(int i=0; i<BUFFER_SIZE; i++){
        sem_wait(&full);
        sem_wait(&mutex);
        item=buffer[out];
        printf("Consumed: %d\n", item);
        out=(out+1)%BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
    }
}

int main(){
    pthread_t p, c;
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);
    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);
    pthread_join(p, NULL);
    pthread_join(c, NULL);
    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);
    return 0;
}
```

```
Activities Terminal Apr 20 17:40
rathnavelu06@localhost:~/rathna/producerConsumer

[rathnavelu06@localhost producerConsumer]$ vi pc.c
[rathnavelu06@localhost producerConsumer]$ gcc pc.c -o pc
[rathnavelu06@localhost producerConsumer]$ ./pc
Producer: 0
Producer: 1
Producer: 2
Producer: 3
Producer: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[rathnavelu06@localhost producerConsumer]$
```